

2(a)	The phase difference is due to the extra distance x travelled by the wave from the student to the wall and back to him ($x = 0.60$ m) and also the phase change of π rad due to reflection at the wall. Phase difference $\phi = \frac{2\pi x}{\lambda} + \pi$ $= \frac{2\pi(0.60)}{0.40} + \pi$ $= 4\pi \text{ rad (or } 0 \text{ or } 2\pi \text{ rad)}$	[1] value of x [1] substitution [1] answer
2(b)(i)	When $x_1 = 1.10$ m, $A_1 = 1.2 \times 10^{-8}$ m When $x_2 = 1.70$ m, $A_2 = ?$ $\frac{A_2}{A_1} = \frac{x_1}{x_2} \rightarrow A_2 = A_1 \left(\frac{x_1}{x_2} \right) = 1.2 \times 10^{-8} \left(\frac{1.10}{1.70} \right)$ $= 0.78 \times 10^{-8} \text{ m}$ Assumption: no energy is absorbed by the wall upon reflection	[1] substitution [1] answer [1] assumption
2(b)(ii)	Since $\phi = 4\pi$ rad, they are in phase, so Amplitude of resultant wave = $(1.2 + 0.78) \times 10^{-8}$ m $= 2.0 \times 10^{-8} \text{ m}$	[1] answer
2(c)	$\frac{I_r}{I_1} = \left(\frac{A_r}{A_1} \right)^2 \rightarrow I_r = I_1 \left(\frac{A_r}{A_1} \right)^2 = 1.0 \times 10^{-6} \left(\frac{2.0 \times 10^{-8}}{1.2 \times 10^{-8}} \right)^2$ $= 2.8 \times 10^{-6} \text{ W m}^{-2}$	[1] substitution [1] answer